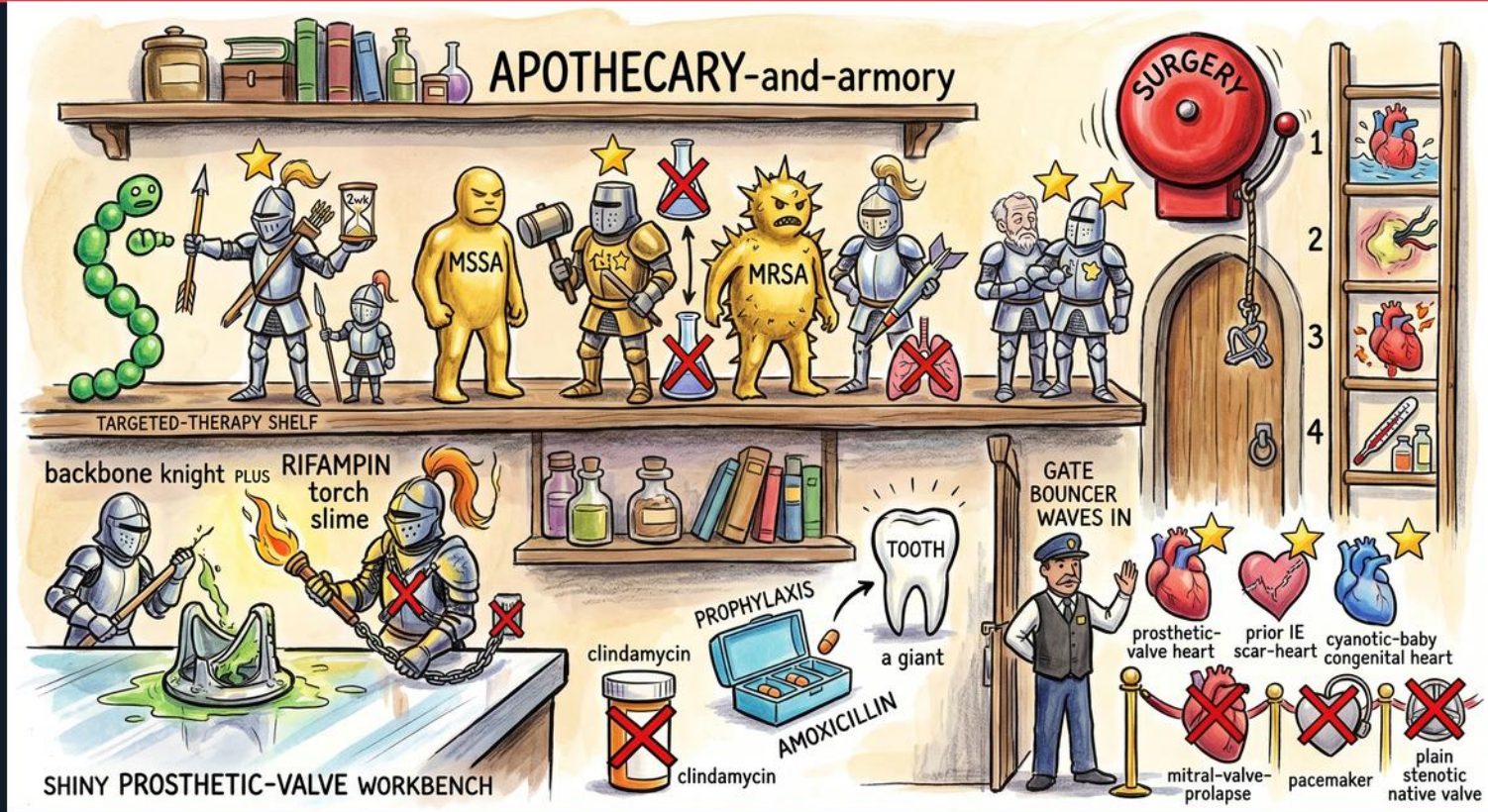


Infective Endocarditis — Treatment, Surgery & Prophylaxis





Strep — penicillin/ceftriaxone (green snake + star)

WHAT YOU SEE

The green serpent-knight with a starred spear and hourglass on the targeted-therapy shelf is penicillin/ceftriaxone for viridans strep, optionally short-coursed.

WHAT IT ENCODES

Penicillin-susceptible viridans strep / *S. gallolyticus* IE: penicillin G or ceftriaxone 2 g IV daily x4 weeks, or a 2-week course with added gentamicin synergy for uncomplicated native-valve disease. Ceftriaxone's once-daily dosing suits outpatient parenteral therapy; penicillin allergy → vancomycin.



Gentamicin = synergy only (small knight)

WHAT YOU SEE

The little squire beside the strep knight is low-dose gentamicin — a synergy helper, never the backbone.

WHAT IT ENCODES

Aminoglycosides are used only for SYNERGY, never as monotherapy: they are not bactericidal alone against strep and only work alongside a cell-wall agent. Gentamicin can shorten susceptible-strep therapy to 2 weeks and is **REQUIRED** for enterococcal cure, but routine gentamicin is **NO LONGER** recommended for native-valve *S. aureus* (nephrotoxicity without benefit).

THE BOARD TRAP

'Add gentamicin to native-valve *S. aureus* for synergy' -> wrong; routine aminoglycoside is no longer recommended in native-valve staph IE — its role is strep dose-shortening and enterococcal synergy, and used alone it is not bactericidal against strep.



MSSA — nafcillin (not vanc)

WHAT YOU SEE

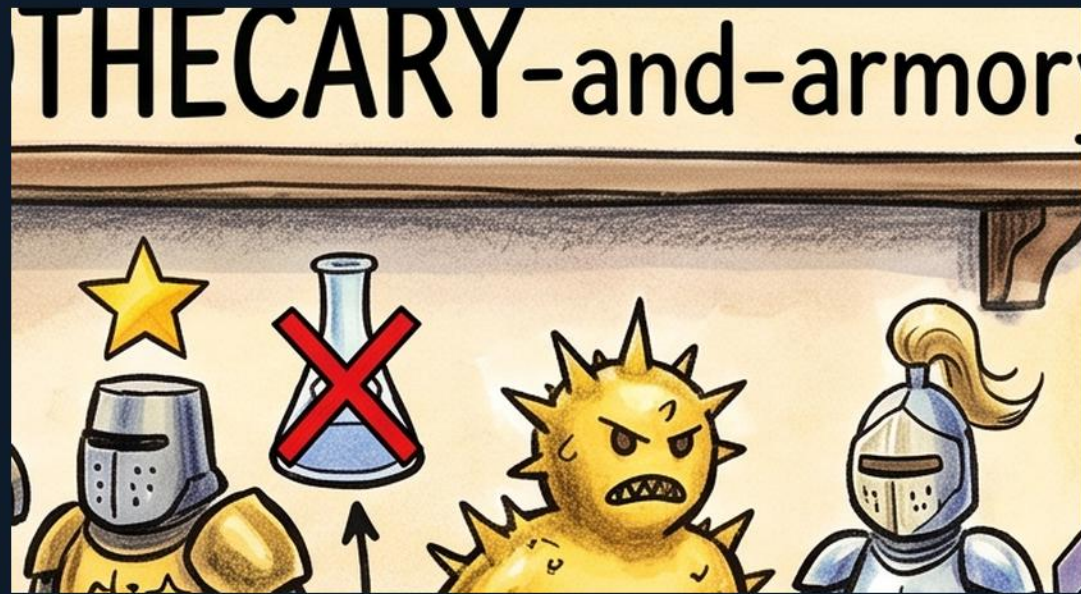
The smooth gold MSSA blob with a crossed-out vancomycin mallet-knight beside it means treat MSSA with nafcillin, not vancomycin.

WHAT IT ENCODES

Methicillin-SUSCEPTIBLE *S. aureus* native valve: nafcillin or oxacillin 2 g IV q4h (or cefazolin 2 g q8h) for 6 weeks. Beta-lactams are bactericidal and clinically SUPERIOR to vancomycin for MSSA (faster clearance, fewer relapses), so empiric vancomycin must be de-escalated once MSSA is confirmed.

THE BOARD TRAP

'Use vancomycin for MSSA IE because it covers staph' -> wrong; vancomycin is INFERIOR to nafcillin/oxacillin/cefazolin for MSSA (slower kill, more failures) — reserve vancomycin for MRSA or true severe beta-lactam allergy, hence the crossed-out mallet.



MRSA — vancomycin/daptomycin (spiky sun)

WHAT YOU SEE

The spiky gold MRSA sun-monster is methicillin-resistant *S. aureus* needing vancomycin or daptomycin.

WHAT IT ENCODES

Methicillin-RESISTANT *S. aureus* native valve: vancomycin (trough 15–20 or AUC/MIC ≥ 400) OR daptomycin (≥ 8 –10 mg/kg IV daily) for 6 weeks. Daptomycin is preferred for vancomycin MIC ≥ 2 or treatment failure; it is inactivated by lung surfactant, so it is fine for right-sided IE with pulmonary emboli but NOT for pneumonia.

THE BOARD TRAP

'If vancomycin fails or the MIC creeps up, just push higher vancomycin doses' -> wrong; switch to daptomycin (or ceftaroline-based salvage) for vancomycin MIC ≥ 2 or persistent MRSA bacteremia — and remember daptomycin is useless for pneumonia due to surfactant inactivation.



Prosthetic valve — backbone + rifampin + gentamicin

WHAT YOU SEE

On the prosthetic-valve workbench the backbone knight wielding a rifampin torch against the biofilm slime is triple therapy for prosthetic staph IE.

WHAT IT ENCODES

Staphylococcal PROSTHETIC-valve endocarditis needs triple therapy: backbone (nafcillin/cefazolin for MSSA, vancomycin for MRSA) PLUS rifampin (penetrates/sterilizes biofilm on foreign material) PLUS gentamicin for the first 2 weeks, with backbone + rifampin continued ≥ 6 weeks. Start rifampin only AFTER bacteremia is controlled (to avoid selecting resistance).

THE BOARD TRAP

'Add rifampin to native-valve *S. aureus*' / 'start rifampin first or alone' -> wrong; rifampin is reserved for PROSTHETIC/foreign-material (biofilm) infection and must be added after the backbone controls bacteremia — it has no role in native-valve staph IE and risks resistance/hepatotoxicity, hence the red X.



Surgery bell — emergent/urgent/elective

WHAT YOU SEE

The red SURGERY alarm bell over a door numbered 1-4 with a ladder of hearts is the tiered surgical-indication call.

WHAT IT ENCODES

Surgical urgency is tiered. EMERGENT (same day): refractory pulmonary edema or cardiogenic shock from acute regurgitation — heart failure is the leading indication and cause of death. URGENT (1-2 days): perivalvular abscess/new heart block, persistent infection despite $\geq 5-7$ days of appropriate antibiotics, large mobile vegetation with recurrent emboli, prosthetic dehiscence, fungal/resistant organisms. ELECTIVE: stable patients with a firm indication who can finish more antibiotics first. Still, $>50\%$ of IE (especially right-sided IVDU) is cured medically.

THE BOARD TRAP

'Delay surgery to finish the full antibiotic course before operating on complicated IE' -> wrong; in HF, abscess, persistent infection, or large embolizing vegetations, early surgery improves survival and remaining antibiotics are given postop — but don't over-call surgery for uncomplicated/right-sided disease, which is usually medical.



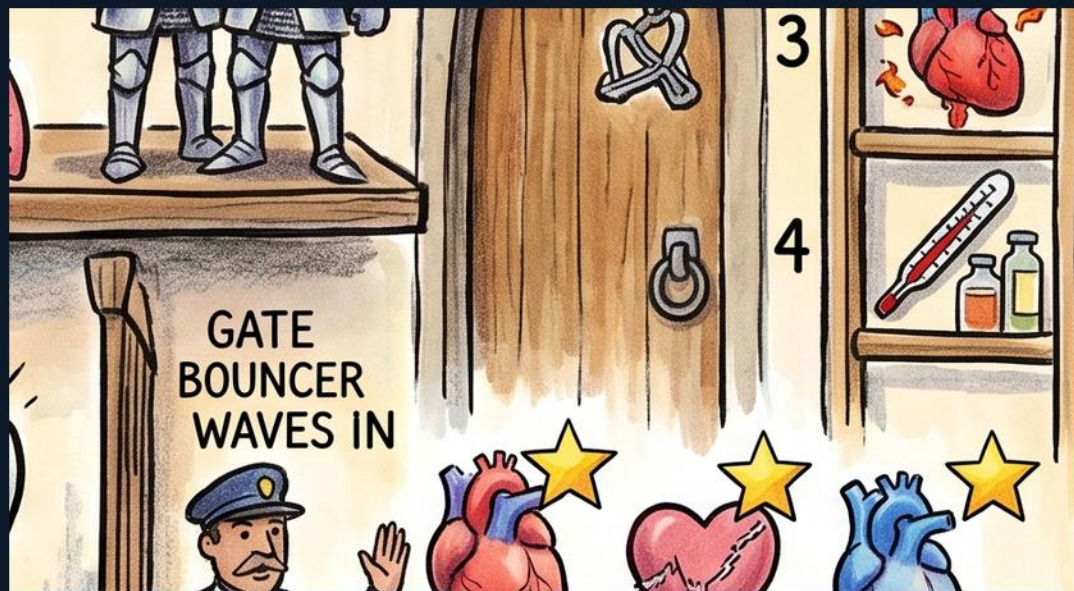
Prophylaxis — highest-risk only (gate bouncer)

WHAT YOU SEE

The uniformed gate bouncer waving certain hearts through is prophylaxis reserved for only the highest-risk cardiac conditions.

WHAT IT ENCODES

Antibiotic prophylaxis is **ONLY** for conditions at highest risk of an adverse IE outcome: (1) prosthetic valve or prosthetic repair material, (2) prior IE, (3) unrepaired cyanotic congenital heart disease, (4) repaired CHD with prosthetic material within the first 6 months, (5) repaired CHD with a residual defect at/near the prosthetic site, (6) cardiac transplant recipients with valvulopathy. Since 2007 (AHA), prophylaxis for the general valve population was dropped — most bacteremia comes from daily activities, not procedures.



Starred hearts admitted (prosthetic / prior IE / cyanotic CHD)

WHAT YOU SEE

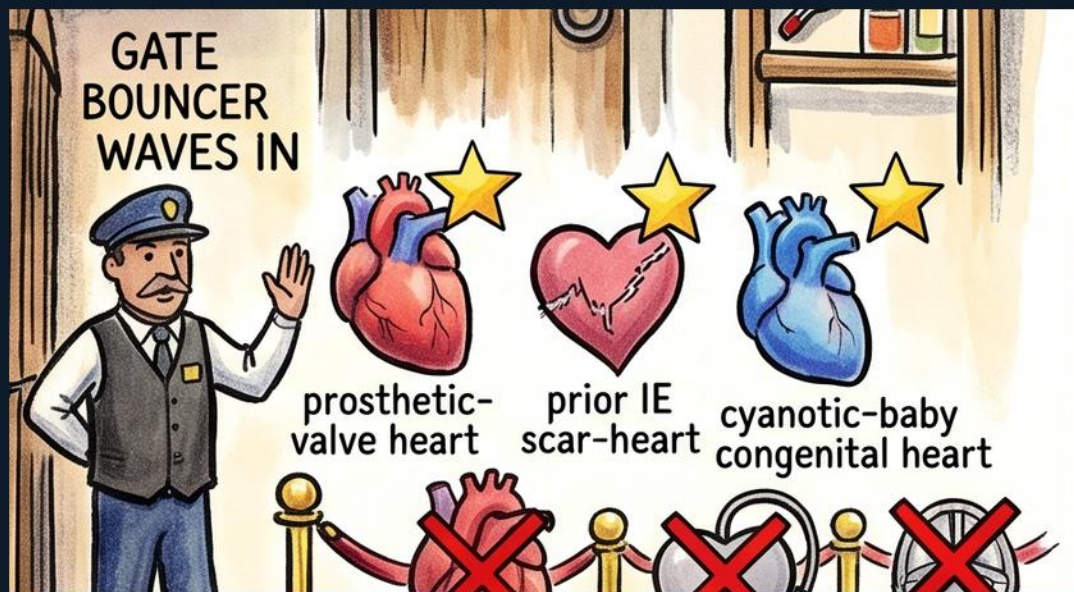
The three gold-starred hearts let through the gate — prosthetic valve, prior-IE scar heart, cyanotic-baby congenital heart — are the high-risk lesions that DO get prophylaxis.

WHAT IT ENCODES

The classic admitted three: ALL prosthetic valve types (mechanical, bioprosthetic, TAVI/TAVR, repair material) — any prosthetic material is a high-risk nidus; PRIOR IE — a permanent indication because it strongly predicts recurrence; and unrepaired CYANOTIC congenital heart disease (TOF, transposition, single-ventricle, palliative shunts). Also on the list: repaired CHD with prosthetic material <6 months, residual defect near a patch, and transplant valvulopathy.

THE BOARD TRAP

'Only mechanical valves (not bioprosthetic/TAVI) get prophylaxis' or 'a recovered prior-IE patient no longer needs it' -> wrong; ALL prosthetic valve types qualify equally, and prior IE is a PERMANENT high-risk indication regardless of how long ago recovery occurred.



Red-X hearts barred (MVP / pacemaker / native AS-MS)

WHAT YOU SEE

The crossed-out hearts the bouncer turns away — mitral-valve-prolapse, pacemaker, plain stenotic native valve — are the lesions that do NOT get prophylaxis.

WHAT IT ENCODES

NOT high-risk, so NO prophylaxis: mitral valve prolapse (even with regurgitation), HCM, an isolated pacemaker/ICD without valve disease, and acquired native-valve disease (mitral stenosis, aortic stenosis, rheumatic disease, bicuspid aortic valve). These were covered pre-2007, which is exactly why they are favorite distractors.

THE BOARD TRAP

'Give amoxicillin before dental work for MVP, HCM, a pacemaker, or rheumatic mitral/aortic stenosis' -> wrong; none of these are on the high-risk list — only prosthetic valves/material, prior IE, specific cyanotic CHD, and transplant valvulopathy qualify. Over-prescribing prophylaxis is the most tested error.



Dental procedures only (giant tooth)

WHAT YOU SEE

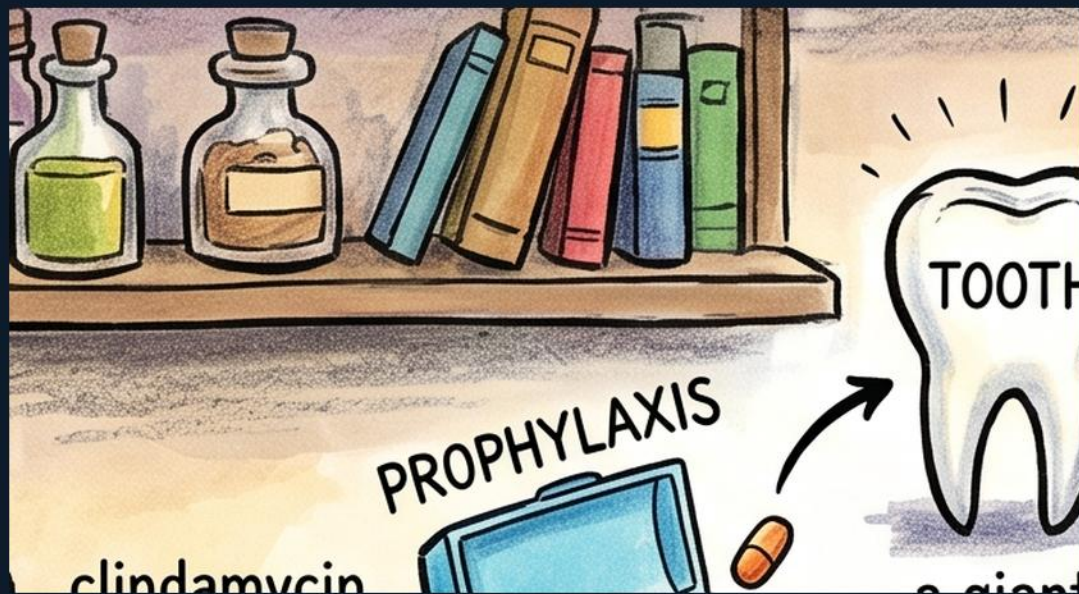
The giant tooth labeled prophylaxis marks that even high-risk patients only get prophylaxis for gingival/periapical dental work.

WHAT IT ENCODES

Even in high-risk patients, prophylaxis is given **ONLY** for procedures manipulating **GINGIVAL** tissue or the **PERIAPICAL** region, or **PERFORATING** oral mucosa (extractions, periodontal surgery, implants, scaling), and certain invasive respiratory-tract procedures. Routine cleanings, simple fillings, and shedding deciduous teeth do **NOT** qualify. GI/GU procedures (colonoscopy, cystoscopy, EGD) do **NOT** require prophylaxis unless there is active infection.

THE BOARD TRAP

'A high-risk patient needs prophylaxis before a screening colonoscopy/cystoscopy, or a routine cleaning' -> wrong; only gingival/periapical/oral-mucosa-perforating dental (and some respiratory) procedures qualify — GI/GU endoscopy and routine cleanings do not unless active infection is present.



Amoxicillin 2 g PO single dose

WHAT YOU SEE

The AMOXICILLIN box on the bench is the single-dose first-line prophylaxis regimen.

WHAT IT ENCODES

First-line prophylaxis: amoxicillin 2 g PO (children 50 mg/kg, max 2 g) as a SINGLE dose 30–60 minutes before the procedure (can be given up to 2 h after if missed). If NPO: ampicillin 2 g IV/IM, or cefazolin/ceftriaxone 1 g IV/IM. Penicillin allergy: cephalexin 2 g PO (avoid if anaphylaxis), azithromycin/clarithromycin 500 mg, or doxycycline 100 mg.

THE BOARD TRAP

'Give a multi-day course of amoxicillin around the procedure' or 'an NPO patient just skips prophylaxis' -> wrong; prophylaxis is a SINGLE pre-procedure dose, and NPO patients switch to a parenteral agent (ampicillin 2 g IV/IM or cefazolin/ceftriaxone 1 g) rather than omitting it.



Clindamycin removed (red-X pills)

WHAT YOU SEE

The two crossed-out clindamycin pills mean clindamycin is no longer a prophylaxis option.

WHAT IT ENCODES

Clindamycin has been REMOVED from IE prophylaxis (2021 AHA update) because of disproportionate *C. difficile* colitis and adverse events. Current penicillin-allergy options are cephalexin (non-anaphylactic), azithromycin/clarithromycin, or doxycycline.

THE BOARD TRAP

'Use clindamycin for penicillin-allergic IE prophylaxis' -> wrong; clindamycin is no longer recommended (*C. diff*/toxicity) — the current allergy alternatives are cephalexin, a macrolide, or doxycycline.

Summary — every symbol

1. Strep — penicillin/ceftriaxone (green snake + star)

Penicillin-susceptible viridans strep / S

3. MSSA — nafcillin (not vanc)

Methicillin-SUSCEPTIBLE S

5. Prosthetic valve — backbone + rifampin + gentamicin

Staphylococcal PROSTHETIC-valve endocarditis needs triple therapy: backbone (nafcillin/cefazolin for MSSA, vancomycin for MRSA) PLUS rifampin (penetrates/sterilizes biofilm on foreign material) PLUS gentamicin for the first 2 weeks, with backbone + rifampin continued ≥ 6 weeks

7. Prophylaxis — highest-risk only (gate bouncer)

Antibiotic prophylaxis is ONLY for conditions at highest risk of an adverse IE outcome: (1) prosthetic valve or prosthetic repair material, (2) prior IE, (3) unrepaired cyanotic congenital heart disease, (4) repaired CHD with prosthetic material within the first 6 months, (5) repaired CHD with a residual defect at/near the prosthetic site,

9. Red-X hearts barred (MVP / pacemaker / native AS-MS)

NOT high-risk, so NO prophylaxis: mitral valve prolapse (even with regurgitation), HCM, an isolated pacemaker/ICD without valve disease, and acquired native-valve disease (mitral stenosis, aortic stenosis, rheumatic disease, bicuspid aortic valve)

11. Amoxicillin 2 g PO single dose

First-line prophylaxis: amoxicillin 2 g PO (children 50 mg/kg, max 2 g) as a SINGLE dose 30–60 minutes before the procedure (can be given up to 2 h after if missed)

2. Gentamicin = synergy only (small knight)

Aminoglycosides are used only for SYNERGY, never as monotherapy: they are not bactericidal alone against strep and only work alongside a cell-wall agent

4. MRSA — vancomycin/daptomycin (spiky sun)

Methicillin-RESISTANT S

6. Surgery bell — emergent/urgent/elective

Surgical urgency is tiered

8. Starred hearts admitted (prosthetic / prior IE / cyanotic CHD)

The classic admitted three: ALL prosthetic valve types (mechanical, bioprosthetic, TAVI/TAVR, repair material) — any prosthetic material is a high-risk nidus; PRIOR IE — a permanent indication because it strongly predicts recurrence; and unrepaired CYANOTIC congenital heart disease (TOF, transposition,

10. Dental procedures only (giant tooth)

Even in high-risk patients, prophylaxis is given ONLY for procedures manipulating GINGIVAL tissue or the PERIAPICAL region, or PERFORATING oral mucosa (extractions, periodontal surgery, implants, scaling), and certain invasive respiratory-tract procedures

12. Clindamycin removed (red-X pills)

Clindamycin has been REMOVED from IE prophylaxis (2021 AHA update) because of disproportionate C